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Personalized Longitudinal Medical Report Generation via Temporally-Aware Federated Adaptation

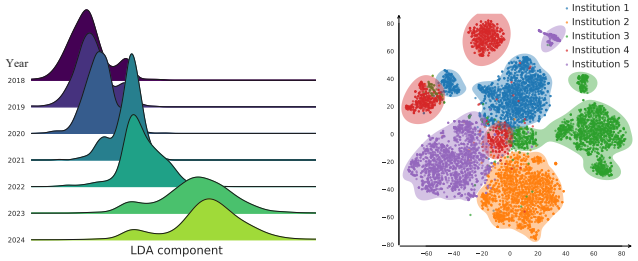
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Abstract

Longitudinal medical report generation is clinically important yet remains challenging due to strict privacy constraints and the evolving nature of disease progression. Although federated learning (FL) enables collaborative training without data sharing, existing FL methods largely overlook longitudinal dynamics by assuming stationary client distributions, making them unable to model temporal shifts across visits or patient-specific heterogeneity—ultimately leading to unstable optimization and suboptimal report generation. We introduce Federated Temporal Adaptation (FTA), a federated setting that explicitly accounts for the temporal evolution of client data. Building upon this setting, we propose FedTAR, a framework that integrates demographic-driven personalization with time-aware global aggregation. FedTAR generates lightweight LoRA adapters from demographic embeddings and performs temporal residual aggregation, where updates from different visits are weighted by a meta-learned temporal policy optimized via first-order MAML. Experiments on J-MID (1M exams) and MIMIC-CXR demonstrate consistent improvements in linguistic accuracy, temporal coherence, and cross-site generalization, establishing FedTAR as a robust and privacy-preserving paradigm for federated longitudinal modeling.[1]

1. Introduction

Automatic medical report generation [19, 26], particularly from longitudinal chest CT scans that capture disease progression [28], is central to reliable clinical diagnosis [16]. By summarizing sequential imaging evidence and modeling temporal patterns across visits, it alleviates the manual reporting burden and improves diagnostic consistency. However, constructing temporally coherent and generalizable models remains challenging due to strict privacy regulations and fragmented institutional data silos [7, 30]. Crucially, these constraints do not only limit the amount of available data, but also break the assumption that a patient's studies can be viewed as i.i.d. samples from a single static distribution. These barriers exacerbate inter-site and temporal heterogeneity, calling for a federated and time-aware setting capable of learning from distributed longitudinal cohorts without compromising privacy.



(a) (b)

Figure 1. Visualization of semantic structure evolution in medical reports. All data are derived from the J-MID dataset. (a) Ridgeline kernel-density plots of LDA-derived topic components for annual reports (2018–2024) from one institution. (b) T-SNE embedding of TF-IDF vectors for reports from five institutions.

Federated Learning (FL) has emerged as a central paradigm for privacy-preserving collaboration across medical institutions [1, 14, 20, 24], yet most formulations remain temporally static, modeling each client as a fixed data distribution. Recent FL approaches to medical report generation, including FedMRG [27], FedMME [38], and Chen et al. [4], follow this setting and treat all examinations on a client as if they were drawn from a single time-invariant distribution, thus ignoring longitudinal disease trajectories and patient-specific evolution. In practice, however, temporal and cross-site drift are pronounced: in our longitudinal chest CT cohort, the semantic distributions of radiology reports shift markedly across years, and study embeddings form well-separated clusters across institutions (Fig. 1). Under such temporal heterogeneity, conventional "one-size-fits-all" aggregation strategies implicitly assume stationarity and tend

[1]Code: https://github.com/zhuhe98/FedTAR-Medical-Report-Generation

to average out progression-related signals that are crucial for clinical interpretation. In other words, existing FL formulations are client-distribution–aware but essentially time-agnostic, leaving the longitudinal nature of medical report generation fundamentally under-modeled and calling for a federated setting that performs personalized, time-aware adaptation to evolving patient states and cross-institutional diversity.

We formalize Federated Temporal Adaptation (FTA), a novel FL problem setting that enables time-aware and personalized modeling under evolving data distributions and privacy constraints. FTA addresses temporal non-stationarity across sequential observations and entity-specific heterogeneity, challenges pervasive in longitudinal, streaming, and cross-institutional domains. To instantiate this setting, we propose FedTAR, a unified framework that integrates demographic-conditioned personalization and meta-learned temporal aggregation for parameter-efficient and theoretically stable optimization. At the personalization level, each client encodes demographic information into compact statistical representations that drive a lightweight hypernetwork to produce patient-specific low-rank adapters. This mechanism provides a parameter-efficient way to capture who varies across federated participants while preserving privacy by avoiding the exchange of raw demographics. At the global level, FedTAR performs temporal residual aggregation, learning visit-dependent weights that balance recent and historical updates. This yields a transferable temporal prior that captures when variation occurs, supporting stable optimization under evolving client distributions. Together, these personalized adapters and temporal aggregation form the first concrete instantiation of FTA, jointly modeling who varies and when change occurs to achieve temporally coherent and privacy-preserving federated report generation.

We use the Japan Medical Image Database (J-MID)[2] [10], which contains $\approx$ 1 M CT/MR exams ($\approx$ 300 M DICOM slices) from more than ten hospitals. The dataset links repeated scans of each patient, providing a unique large-scale resource for evaluating both cross-site federated and longitudinal models. We also validated the framework on public datasets MIMIC-CXR [9], demonstrating its robustness across diverse clinical domains.

Our primary contributions can be summarized as follows:

- We introduce Federated Temporal Adaptation (FTA), a new FL setting that makes temporal evolution a first-class component of federated optimization.
- We present FedTAR, the first instantiation of FTA, unifying personalized low-rank adapters with temporal residual aggregation for stable and efficient learning under evolving client distributions.
- Experiments on J-MID and MIMIC-CXR demonstrate improved report quality, temporal coherence, and cross-site generalization, supported by convergence analysis.

[2] https://www.radiology.jp/j-mid/

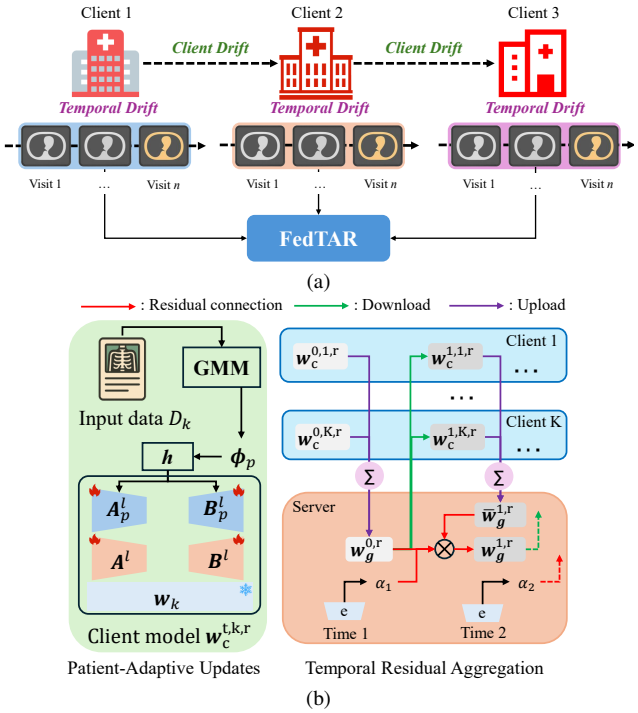


Figure 2. Overview of the problem and our approach. (a) Client- and temporal-drift challenges in federated longitudinal medical report generation. (b) The proposed FedTAR framework.

2. Related Work

2.1. Federated Learning for Medical Applications

Federated Learning (FL) [3, 14, 33] enables multi-institutional model training without data sharing [23]. Beyond early classification tasks, recent studies explored generative objectives such as medical report synthesis. Chen and Pan [5] combined FedAvg with evidence-driven optimization to improve global modeling, while FedMRG [27] mitigated inter-client heterogeneity but assumed static data. FedMME [37] exploited large vision–language models for single-round FL, focusing on non-Independent and Identically Distributed (non-IID) robustness rather than temporal evolution. Parameter-efficient personalization methods like PerAda [39] and pFedLoRA [40] tailor local adapters yet remain time-invariant. Collectively, these works advance multimodal federated generation but overlook temporal non-stationarity and longitudinal patient dynamics—the core focus of our Federated Temporal Adaptation setting.

2.2. Temporal Longitudinal Medical Imaging

Longitudinal imaging is vital for tracking disease progression and guiding diagnosis [12, 17], yet early report generation models overlooked temporal dependencies by treating

each scan independently. Hamamci et al. [15] introduced CT2Rep and CT2RepLong, which employ memory-driven decoders and cross-attention to fuse multi-timepoint CTs and prior reports, improving temporal coherence and diagnostic fidelity. Though centralized, these works motivate privacy-preserving longitudinal modeling in federated settings. In parallel, Ghari and Shen [13] enhanced streaming robustness via checkpoint ensembling to counter concept drift [11], while Chen et al. [6] proposed time-weighted aggregation for evolving data. Together, these advances highlight the need for temporal fusion [25] and dynamic optimization [2], whose integration into federated medical report generation remains largely unexplored.

3. Methodology

3.1. FTA Problem Definition

We consider Federated Temporal Adaptation (FTA) as a federated learning setting where each client holds sequential data collected over multiple time steps. Formally, client $k \in \{1, \dots, K\}$ possesses a local dataset $\mathcal{D}_k = \{p_n\}_{n=1}^{n_k}$, and each sequence

$$p_n = \{x_{k,t}, y_{k,t}\}_{t=1}^{T}, (x_{k,t}, y_{k,t}) \sim P_{k,t},$$

contains the input $x_{k,t}$ observed at time step t and the corresponding target $y_{k,t}$ drawn from a time-varying client-specific distribution $P_{k,t}$. The global objective is to learn a model that performs well across both clients and time, typically written as

$$\min_{\boldsymbol{w}} \sum_{k=1}^{K} \sum_{t=1}^{T} \mathcal{L}\big(f(\boldsymbol{w}; x_{k,t}),\, y_{k,t}\big), \tag{1}$$

where $\boldsymbol{w}$ denotes the global model parameters, $f(\cdot; \boldsymbol{w})$ is the prediction model, and $\mathcal{L}$ is a task-specific loss function.

Compared with standard FL, which assumes each client k draws i.i.d. samples $(x, y) \sim P_k$ from a stationary distribution, and online FL, which receives streaming mini-batches $\mathcal{B}_{k,\tau} \sim Q_{k,\tau}$ over rounds τ where the time index is attached to the optimization process rather than the data distribution, FTA explicitly models client-wise temporal distributions $\{P_{k,t}\}_{t=1}^{T}$ with sequences $\{(x_{k,t}, y_{k,t})\}_{t=1}^{T}$, and thus making temporal drift a first-class component of the federated objective. The learning objective is therefore to obtain a global model that adapts coherently across time while remaining consistent across clients under privacy constraints. A more formal comparison between FTA and other FL problem definitions is provided in the Appendix. In this work, we instantiate this general formulation in the context of longitudinal medical report generation, where sequential imaging and textual records are distributed across institutions.

3.2. Preliminary

An overview of our method is shown in Fig. 2. We propose a novel multimodal FL framework consists of two sequentially coupled modules: (1) Client-Side Patient-Adaptive Updates, where we encode low-dimensional patient data via GMMs and dynamically generate personalized LoRA weights for local fine-tuning; and (2) Temporal Residual Aggregation, which computes residuals at each temporal step and employs hypergradient-based meta-learning to adaptively weight and aggregate client updates. We further integrate a hierarchical communication protocol to minimize overhead. Consequently, we deliver both theoretical soundness and practical efficiency for longitudinal medical report generation.

3.3. Client-Side Patient-Adaptive Updates

To personalize local adaptation, we condition LoRA weight generation on patient-specific data. Each client k receives global parameters w and its patient demographics $\text{info}_k = \{\text{id}_p, \text{age}_p, \text{sex}_p\}_{p=1}^{n_k}$. We compute a normalized profile vector $\boldsymbol{v}_p \in \mathbb{R}^3$ for each patient p as:

$$\boldsymbol{v}_p = \big[\text{Norm}(\text{hash}(\text{id}_p)), \text{Norm}(\text{age}_p), \text{enc}(\text{sex}_p)\big]^\top. \tag{2}$$

Then, we encode $\boldsymbol{v}_p$ into a soft cluster assignment $\mathbf{q}_p$ via a GMM as $\mathbf{q}_p = \text{GMM}(\boldsymbol{v}_p) \in \Delta^{n_{comp}-1}$. The soft assignment captures the probabilistic membership of $\boldsymbol{v}_p$ across n_{comp} latent clusters, allowing for smooth interpolation between different patient subgroups. Subsequently, the probability vector $\mathbf{q}_p$ is linearly projected into a d-dimensional embedding space through a learnable transformation, yielding the patient embedding $\boldsymbol{\phi}_p = \mathbf{W}_{\text{proj}}\mathbf{q}_p + \mathbf{b}_{\text{proj}} \in \mathbb{R}^d$. This projection enables the model to adaptively reweight cluster contributions based on downstream tasks.

Thereafter, we use each patient embedding $\boldsymbol{\phi}_p$ to modulate a low-rank adapter. Let $W^l_{\text{client}} \in \mathbb{R}^{e\times e}$ be the l-th client model weight block, and suppose our client LoRA adapter is

$$\Delta W^l_{\text{client}} = A^l B^{l\top}, \quad A^l \in \mathbb{R}^{e\times r},\ B^l \in \mathbb{R}^{r\times e}.$$

We introduce a lightweight hypernetwork $h^l : \mathbb{R}^d \to \mathbb{R}^{(e\times r)+(r\times e)}$ that maps $\boldsymbol{\phi}_p$ to patient-specific adapter parameters as

$$\big[A^l_p(\boldsymbol{\phi}_p),\, B^l_p(\boldsymbol{\phi}_p)\big] = h^l(\boldsymbol{\phi}_p), \tag{3}$$

and define the total adapted weight for patient p at layer l as follows:

$$W^l_p = W^l_{\text{client}} + \underbrace{A^l B^{l\top}}_{\Delta W^l_{\text{client}}} + \underbrace{A^l_p(\boldsymbol{\phi}_p)\, B^l_p(\boldsymbol{\phi}_p)^\top}_{\Delta W^l_p}. \tag{4}$$

At each forward pass for client k, we retrieve its demographics set $\{p \in D_k\}$, compute each $\boldsymbol{\phi}_p$, instantiate $\{\Delta W^l_p\}$ via the hypernetworks, and apply W^l_p in place of W^l_{client}.

Training proceeds by minimizing the standard clinical loss

$$\mathcal{L}\big(\{W^l_{\text{client}}\}, \{A^l, B^l\}, \{h^l\}\big) \;=\; \sum_{(I,R)\in\mathcal{D}_k} \ell\big(f(I; W_p),\, R\big), \tag{5}$$

where gradients flow into the client model weights W^l_{client}, the shared adapter factors $\{A^l, B^l\}$, the hypernetwork parameters $\{h^l\}$, and the projection of $\boldsymbol{\phi}_p$.

Overall, our client-side patient-adaptive updates integrate demographic-conditioned soft clustering with hypernetwork-based LoRA, enabling fine-grained personalization with minimal communication overhead. This design allows smooth interpolation across latent subpopulations and improves generalization under heterogeneous clinical data (Algorithm 1).

Algorithm 1 Client-Side Patient-Adaptive Updates

Require: LoRA adapter $\{A^l, B^l\}_{l=1}^{L}$, hypernetworks $\{h^l\}_{l=1}^{L}$, projection params $(\mathbf{W}_{\text{proj}}, \mathbf{b}_{\text{proj}})$, local data (I, R)

for all patients $p \in D_k$ **do**
  $\boldsymbol{v}_p \leftarrow \big[\text{Norm}(\text{hash}(\text{id}_\text{p})),\ \text{Norm}(\text{age}_\text{p}),\ \text{enc}(\text{sex}_\text{p})\big]^\top$
  $\mathbf{q}_p \leftarrow \text{GMM}(\boldsymbol{v}_p)$; $\boldsymbol{\phi}_p \leftarrow \mathbf{W}_{\text{proj}}\,\mathbf{q}_p + \mathbf{b}_{\text{proj}}$
  for $l = 1$ to L **do**
    $(A^l_p, B^l_p) \leftarrow h^l(\boldsymbol{\phi}_p)$
    $\Delta W^l_{\text{client}} \leftarrow A^l\, B^{l\top}$; $\Delta W^l_p \leftarrow A^l_p\, B^{l\top}_p$
    $W^l_p \leftarrow W^l_{\text{client}} + \Delta W^l_{\text{client}} + \Delta W^l_p$
  end for
end for
Compute $\mathcal{L} \;=\; \sum_{(I_p,R_p)\in D_k} \ell\big(f((I_p; \{W^l_p\}), R_p)\big)$
Update $\{A^l, B^l\}, \{h^l\}, \mathbf{W}_{\text{proj}}, \mathbf{b}_{\text{proj}}$ via backprop

3.4. Temporal Residual Aggregation

Longitudinal clients continuously accumulate new visits, causing their optimization trajectories to drift in both direction and speed; because static parameter averaging cannot anticipate or align these future temporal shifts, we aggregate residual updates, and use meta-learning to adapt their weights so the global model can predictively adjust to fast-changing, temporally expanding data.

3.4.1. Two Layer Temporal Aggregation

At communication round r, each client k sequentially uploads its locally fine-tuned models $\{\boldsymbol{w}_c^{t,k,r}\}_{t=0}^{T}$ after training on time-point t data. At time step t, we perform a single round of local weighted model aggregation to obtain the base global model weights $\bar{\boldsymbol{w}}_g^{(t,r)}$:

$$\bar{\boldsymbol{w}}_g^{(t,r)} = \sum_{k=1}^{K} \frac{1}{n_k} \boldsymbol{w}_c^{t,k,r}\,. \tag{6}$$

Subsequently, rather than discarding previous temporal information, we incorporate a residual correction as follows:

$$\boldsymbol{w}_g^{(t,r)} = \boldsymbol{w}_g^{(t-1,r)} + \alpha_t\big(\bar{\boldsymbol{w}}_g^{(t,r)} - \boldsymbol{w}_g^{(t-1,r)}\big), \tag{7}$$

where we initialize $\boldsymbol{w}_g^{(0,r)} = \boldsymbol{w}_g^{(T,r-1)}$ for $r > 1$ and $\boldsymbol{w}_g^{(0,1)}$ is set to the initial global model. Thus, at each time step, the new global model is obtained by adding a weighted residual that reflects the update magnitude at that time point.

The per-step coefficient $\alpha_t \in [0, 1]$ will be learned via a meta-learning procedure. However, even before specifying a particular schedule for $\{\alpha_t\}$, the following theorems reveal intrinsic convergence and stability properties of the residual formulation, thereby justifying the subsequent meta-learning of α_t.

Theorem 1. *Let* $\Delta_t = \bar{\boldsymbol{w}}_g^{(t,r)} - \boldsymbol{w}_g^{(t-1,r)}$, $\beta_x^{(t)} = \alpha_x \prod_{y=x+1}^{t}(1-\alpha_y) \quad (1 \le x \le t)$, *and* $\beta_0^{(t)} = \prod_{y=1}^{t}(1-\alpha_y)$. *Then the iterates defined by* (7) *admit the closed form*

$$\boldsymbol{w}_g^{(t,r)} = \beta_0^{(t)}\, \boldsymbol{w}_g^{(0,r)} + \sum_{x=1}^{t} \beta_x^{(t)}\, \bar{\boldsymbol{w}}_g^{(x,r)},$$

with $\beta_x^{(t)} \ge 0$ *and* $\sum_{x=0}^{t} \beta_x^{(t)} = 1$. *Consequently,* $\boldsymbol{w}_g^{(t,r)}$ *always lies in the convex hull of* $\{\bar{\boldsymbol{w}}_g^{(1,r)}, \ldots, \bar{\boldsymbol{w}}_g^{(t,r)}\}$, *and if* $\lim_{t\to\infty} \bar{\boldsymbol{w}}_g^{(t,r)} = w^*$ *then* $\lim_{t\to\infty} \boldsymbol{w}_g^{(t,r)} = w^*$.

This result guarantees that, irrespective of how the meta-learner chooses α_t, the global model retains a smooth memory of all past snapshots by forming a convex combination of $\{\bar{\boldsymbol{w}}_g^{(1,r)}, \ldots, \bar{\boldsymbol{w}}_g^{(t,r)}\}$, where the pre-trained initial model $\boldsymbol{w}_g^{(0,r)}$ serves as an anchoring vertex in the convex hull, thus imbuing the aggregation process with domain-specific priors that are smoothly incorporated and subsequently attenuated according to the learned $\{\alpha_t\}$. When generating medical reports from a patient's longitudinal examination records—our aggregation automatically balances contributions from each time point, ensuring that early biomarkers and recent observations are both reflected in a coherent, temporally aware global model. A complete analysis is deferred to the Appendix A.1.

Theorem 2. *Assume that for all* t, $\|\Delta_t\| \le G$ *for some constant* $G > 0$, *and that* $0 \le \alpha_t \le 1$. *Then the update magnitude at each time step satisfies*

$$\big\|\boldsymbol{w}_g^{(t,r)} - \boldsymbol{w}_g^{(t-1,r)}\big\| = \alpha_t \,\|\Delta_t\| \;\le\; \alpha_t\, G.$$

In particular, if $\alpha_t \to 0$ *as* $t \to \infty$, *then the per-step update norm vanishes, ensuring smoother convergence and preventing excessive oscillations.*

This boundedness property underpins the trade-off between adaptivity and stability: large α_t allows rapid response to new data, whereas small α_t dampens noise and enforces convergence. Consequently, meta-learning $\{\alpha_t\}$ enables an automated balance between optimization speed and robust stability in non-stationary federated settings. A complete analysis is deferred to the Appendix A.2.

Algorithm 2 Temporal Residual Aggregation

Require: Initial global model $\boldsymbol{w}_g^{(0,0)}$, meta-parameters ψ, total rounds R, time points T

for $r = 1$ to R **do**
  $\boldsymbol{w}_g^{(0,r)} \leftarrow \boldsymbol{w}_g^{(T,\,r-1)}$
  for $t = 1$ to T **do**
    for each client $k = 1, \ldots, K$ in parallel **do**
      $\boldsymbol{w}_c^{t,k,r} \leftarrow \texttt{ClientUpdates}(\boldsymbol{w}_g^{(t-1,r)}, D_k)$
    end for
  $\bar{\boldsymbol{w}}_g^{(t,r)} \leftarrow \sum_{k=1}^{K} \frac{1}{n_k} \boldsymbol{w}_c^{t,k,r}$
  $\boldsymbol{w}_g^{(t,r)} \leftarrow \boldsymbol{w}_g^{(t-1,r)} + \alpha_t(\psi)\,(\bar{\boldsymbol{w}}_g^{(t,r)} - \boldsymbol{w}_g^{(t-1,r)})$
  end for
  $\psi \leftarrow \psi - \eta\,\nabla_\psi\,\mathcal{L}_{\text{val}}(\boldsymbol{w}_g^{(T,r)}(\psi))$
end for

3.4.2. Meta-Learned Coefficients

Building on the convex-combination and bounded-update properties established in Theorems 1 and 2, we introduce a data-driven procedure to calibrate the temporal weights $\{\alpha_t\}$ so as to optimize both responsiveness to recent updates and overall convergence stability. Specifically, we parameterize α_t as

$$u_t = g\big(e(t);\psi\big), \qquad [\alpha_1, \ldots, \alpha_T] = \mathrm{Softmax}([u_1, \ldots, u_T]), \tag{8}$$

where $g(\cdot;\psi)$ denotes a lightweight multilayer perceptron mapping time embeddings $e(t)$ to unnormalized scores. This construction ensures that $\alpha_t \in (0,1)$ and that the resulting residual aggregation remains within the convex hull of past snapshots, thereby preserving the theoretical guarantees of smooth convergence.

However, selecting ψ directly to minimize training loss can lead to overfitting temporal fluctuations; therefore, we adopt a bilevel optimization strategy. In the inner loop, temporal aggregation across $t = 1, \ldots, T$ yields the aggregated model $\boldsymbol{w}_g^{(T,r)}(\psi)$ according to (7). In the outer loop, we update the meta-parameters by minimizing a held-out validation loss:

$$\psi \leftarrow \psi - \eta\,\nabla_\psi\,\mathcal{L}_{\text{val}}\big(\boldsymbol{w}_g^{(T,r)}, \psi\big), \tag{9}$$

where $\mathcal{L}_{\text{val}}$ measures generalization performance on longitudinal medical report generation tasks. The detailed derivation of the hypergradient computation is provided in the Appendix A.3. By leveraging hypergradient descent, this meta-learning scheme systematically refines $\{\alpha_t\}$ to balance optimization speed against robust stability across temporal domains. We summarize this process in Algorithm 2.

Moreover, the softmax parameterization coupled with hypergradient-based adaptation circumvents the need for manual scheduling of residual coefficients, thereby automating the trade-off between rapid adaptation to non-stationary data and dampening of noisy updates. This approach ensures that large coefficients are assigned to time points with significant distributional shifts, whereas minor or spurious fluctuations are attenuated, leading to improved convergence properties and better generalization under temporal heterogeneity. We note that, under standard smoothness and Lipschitz continuity assumptions on the meta-learner, the bilevel optimization procedure admits convergence guarantees; a complete analysis is deferred to the Appendix A.4.

4. Experiments

The objectives of our experimental study are to demonstrate that our temporally-aware federated adaptation framework can effectively capture longitudinal disease progression across five distinct clinical sites, to show that patient demographics–conditioned low-rank adaptation via Gaussian Mixture Modeling enhances the quality of generated chest CT reports compared to conventional FL baselines, and to verify that meta-learned temporal residual aggregation improves convergence stability and generalization performance under heterogeneous institutional data distributions.

Dataset Composition We conducted all experiments on a proprietary, longitudinal chest CT dataset newly assembled from five independent medical institutions, comprising 36,674, 5,967, 6,631, 6,688, and 1,439 patient cases, respectively. Each institution contributed de-identified CT volumes, corresponding radiological narratives authored by board-certified clinicians, and minimal patient demographics—namely a pseudonymized identifier, age, and sex. Crucially, every patient in our cohort underwent exactly five sequential CT examinations, thereby capturing fine-grained temporal progression of pulmonary findings. We also conducted evaluations on the public MIMIC-CXR dataset [9], which contains 370,000 medical images.

Federated Client Partitioning To simulate a realistic FL environment under non-IID conditions, we partitioned the entire dataset by source institution into five clients. Each client holds all five time-ordered examinations for its local patient subpopulation, which enables us to evaluate both the convergence behavior and generalization performance of our patient-adaptive LoRA embeddings and meta-learned temporal aggregation under heterogeneous temporal dynamics. Moreover, by preserving each client's unique distribution of disease trajectories, our setup rigorously tests the ability of Gaussian Mixture–conditioned embedding and hypernetwork-driven adaptation to reconcile local variability with global model consistency. For MIMIC-CXR dataset, we partitioned the data into four clients. Since patient-level identifiers are unavailable, we assigned "subject id" as input.

Table 1. Comparison of our proposed method (FedTAR) with standard FL baselines on the chest CT report generation task. Evaluation metrics include BLEU-n (n=1–4), ROUGE-L, and CIDEr, which collectively reflect linguistic precision, recall, and content relevance. FedTAR achieves consistent improvements, demonstrating the effectiveness of demographics- and temporally-aware adaptation.

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CIDEr
FedAvg [24]	38.40	24.68	16.16	10.98	28.54	31.70
FedProx [21]	38.32	24.48	15.95	11.00	28.58	31.62
SCAFFOLD [18]	35.40	22.38	14.80	10.10	27.12	24.82
FedAdam [31]	38.26	24.54	16.05	10.93	28.61	30.62
FedYogi [31]	35.39	22.37	13.72	10.46	27.91	24.79
DRFA [8]	36.80	24.02	16.63	11.60	28.75	29.51
FedTAR (ours)	**40.08**	**25.94**	**17.80**	**12.40**	**29.54**	**42.80**

Table 2. Ablation study evaluating the contributions of the Gaussian Mixture Model (GMM)-based demographics embedding and the temporal residual aggregation module. Removing either component leads to consistent performance degradation across all metrics, confirming their necessity for capturing patient-specific heterogeneity and longitudinal dynamics.

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CIDEr
FedTAR w/o GMM	**40.38**	25.08	17.59	12.07	28.65	41.56
FedTAR w/o temporal	38.83	25.82	17.28	11.21	29.50	41.88
FedTAR	40.08	**25.94**	**17.80**	**12.40**	**29.54**	**42.80**

Ethics and Privacy All data collection and processing procedures received approval from the respective institutional review boards, and no direct or quasi-identifiers are retained at any stage. However, due to strict patient-privacy regulations, the dataset itself is not publicly released. This configuration provides a robust testbed for assessing optimization stability and personalized generalization in temporally aware federated medical report generation.

4.1. Implementation Details

Our method is implemented in PyTorch 2.4.1 and executed on NVIDIA RTX A6000 GPUs. The image encoder is a Convolutional vision Transformer warmed up with an ImageNet-21K checkpoint, and the text decoder is DistilGPT2 [32] pretrained on clinical corpora. We attach LoRA adapters of rank $r = 4$ with $\alpha = 128$ to every transformer layer and optimize all parameters using AdamW with a base learning rate of 10^{-5} for model and adapter parameters and 10^{-4} for meta-learned temporal coefficients. We implement patient-profile clustering by fitting a diagonal-covariance GMM with sixteen components using scikit-learn's `GaussianMixture` class, configured with a small regularization on the diagonal entries to prevent singularities and a strict convergence tolerance for EM iterations. Each patient is represented by a normalized three-dimensional vector—comprising a SHA-256–based hashed identifier scaled to $[0, 1]$, age divided by 100, and a binary encoding of sex. After fitting the GMM on the stacked matrix of these profile vectors, we obtain for each patient a soft assignment vector of length sixteen via `predict_proba`. These mixture weights are then mapped into the model's embedding space through a learnable linear projection layer, where only the projection's weights and bias are updated during downstream federated training.

The temporal residual aggregation hypernetwork is a lightweight, fully differentiable module that embeds each integer time index via a learnable table, feeds the embeddings into a single-hidden-layer MLP with ReLU to produce unnormalized scores over all time steps, and applies a softmax to obtain aggregation coefficients that sum to one for the residual update.

4.2. Results

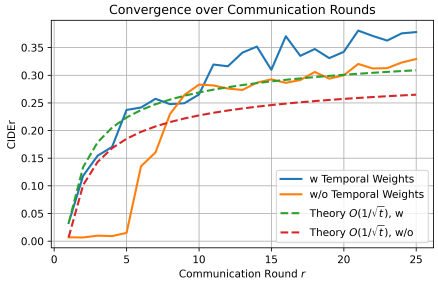
Comparison with Baselines We report BLEU-1 through BLEU-4 [29], ROUGE-L [22], and CIDEr [36] for evaluation [34]. We compare our proposed method (FedTAR) against six representative FL baselines: FedAvg [24], FedProx [21], SCAFFOLD [18], FedAdam [31], FedYogi [31], and DRFA [8]. For all six federated baselines, we follow the conventional centralized-training protocol that pools data from all time points into a single dataset. The average scores across institutions are summarized in Table 1, and detailed per-site results are reported in Appendix B.1. FedTAR consistently surpasses all baselines across BLEU, ROUGE-L, and CIDEr metrics, confirming its advantage in both linguistic precision and semantic consistency. These improvements stem from the framework's ability to condition model updates on patient demographics and temporal progression, allowing the global model to maintain coherence across visits. In particular, FedTAR shows higher early n-gram accuracy

Table 3. Quantitative evaluation on the MIMIC-CXR dataset. Higher values indicate better performance.

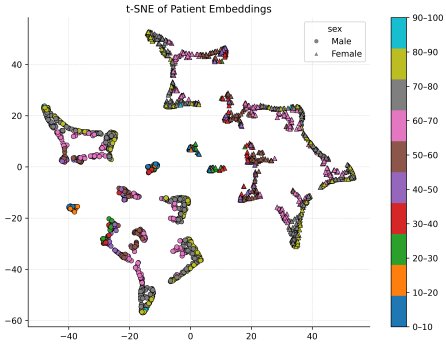
	CE-P	CE-R	CE-F1	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CIDEr
FedAvg [24]	68.58	22.63	28.43	29.16	19.49	14.36	11.00	35.77	37.54
FedProx [21]	**69.83**	24.59	30.26	27.16	17.80	12.85	9.70	34.11	37.38
SCAFFOLD [18]	44.58	22.6	21.47	28.00	17.23	14.21	9.86	38.05	99.11
FedAdam [31]	58.74	25.95	31.80	**39.17**	**28.56**	22.42	18.39	43.08	97.07
FedYogi [31]	52.02	23.36	29.90	35.55	22.57	18.94	13.84	37.03	94.11
DRFA [8]	47.67	26.66	30.30	36.81	26.94	21.30	17.64	42.71	108.10
FedTAR (ours)	52.73	**28.82**	**33.21**	36.59	27.83	**22.60**	**19.08**	**44.42**	**120.58**

Table 4. Comparison with centralized generation models on the J-MID dataset. Higher values are better.

	CE-P	CE-R	CE-F1	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CIDEr
CT2RepLong	13.37	**13.12**	13.24	**47.20**	**28.46**	**24.26**	10.57	35.63	25.68
FedTAR (ours)	**17.65**	12.76	**14.47**	40.08	25.94	20.54	**12.14**	**29.54**	**42.80**



(a) Empirical and theoretical convergence of CIDEr scores over communication rounds.



(b) Visualization of patient embeddings ϕ_p.

Figure 3. Overall caption describing both subfigures.

and stronger structural alignment, indicating its capacity to capture subtle longitudinal evolution and site-specific variations. Qualitative report samples illustrating these gains are presented in Appendix B.2.

Ablation Study As shown in Table 2, we perform ablation experiments on the two core components of FedTAR: the GMM-based demographics embedding and the meta-learned temporal aggregation. Removing the GMM module markedly reduces BLEU and CIDEr scores, confirming that patient-conditioned embeddings are essential for capturing demographic and inter-patient variability. Disabling the temporal hypernetwork similarly lowers ROUGE-L and recall-oriented metrics, demonstrating that adaptive temporal weighting is crucial for modeling longitudinal disease progression and mitigating inter-site drift. These complementary effects verify that both personalization and temporal adaptation are indispensable to the overall performance. Extended hyperparameter sensitivity to the LoRA rank r and GMM component number k is reported in Appendix B.3.

Computational and Communication Efficiency. We evaluate the overhead introduced by FedTAR. Adding LoRA adapters and the GMM-based demographics encoder increases per-layer FLOPs by less than 0.001% on average, and GMM fitting is performed once per client, adding $< 0.5\%$ total runtime. During communication, only lightweight LoRA parameters (A, B) and their gradients are exchanged—no patient-specific tensors or demographics leave the client—reducing payload size by over $10\times$ compared with full model synchronization. Overall, the proposed modules deliver substantial performance gains with negligible additional computational or communication cost.

Cross-dataset Evaluation We further validated the framework on the MIMIC-CXR dataset using the same settings as in the main study. Clinical Efficacy (CE) metrics [35], which measure clinical consistency between generated and reference reports, were employed alongside standard text metrics. As shown in Table 3, our method consistently outperforms prior approaches, achieving relative gains of +4.4% in macro-F1, +8.1% in macro-Recall, and +11.5% in CIDEr,

Table 5. Paired significance analysis between FedTAR and FedAdam on MIMIC-CXR. Significant results: 95% CI excludes 0 and $p < 0.05$.

Statistic	CE-P	**CE-R**	**CE-F1**	BLEU-1	**BLEU-2**	**BLEU-3**	**BLEU-4**	**CIDEr**	**ROUGE-L**
95% CI	(-0.02,0.08)	(0.01,0.10)	(0.00,0.09)	(-0.00,0.02)	(0.01,0.03)	(0.01,0.03)	(0.01,0.03)	(0.17,0.57)	(0.01,0.03)
p-value	0.25	0.02	0.04	0.09	<0.001	<0.001	<0.001	<0.001	<0.001
Win-rate (%)	88.98	98.84	97.76	95.71	100.00	100.00	100.00	100.00	100.00
Significant	No	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes

along with improvements in ROUGE-L and BLEU-4. These results demonstrate stronger semantic fidelity and clinical consistency in generated reports.

Comparison with Centralized Report Generation To evaluate cross-site generalization, we compared our federated framework with centralized report generation models CT2RepLong on the J-MID dataset. As shown in Table 4, our method achieves higher macro-F1, CE precision, and a 66.7% gain in CIDEr, indicating stronger semantic alignment and phase-level coherence. Despite a slight recall drop (2.7%), the overall results demonstrate that the proposed temporal residual aggregation effectively preserves privacy while maintaining or surpassing centralized performance across key clinical and semantic metrics.

Statistical Significance Analysis We validate that the observed improvements are statistically reliable by comparing FedTAR with the strongest baseline FedAdam on the MIMIC-CXR dataset. For each study, we computed the metric difference and drew 5000 bootstrap replicates to estimate 95% confidence intervals (CI). Paired permutation tests provided p-values, and the win-rate indicates the percentage of test studies where FedTAR outperforms FedAdam. A metric is significant when the 95% CI excludes 0 and $p < 0.05$.

Convergence Analysis Figure 3a plots the empirical CIDEr trajectories of temporal residual aggregation, its unweighted variant, and the $O(\frac{1}{\sqrt{t}})$ reference. FedTAR increases rapidly in the early rounds and saturates before the baseline shows noticeable gains. Across training, the blue curve stays above the theoretical reference with reduced variance, indicating faster and more stable convergence. This supports our design: demographics-conditioned LoRA adapters align updates within patient subspaces, while the temporal hypernetwork prioritizes recent rounds without amplifying noise, thereby accelerating learning and mitigating heterogeneity-induced oscillations.

Impact of GMM The GMM-based demographics embedding allows the model to capture fine-grained patient heterogeneity via soft cluster assignments, producing ϕ_p that reflect subtle demographic and disease progression differences. This yields higher CIDEr and ROUGE scores. Figure 3b shows a t-SNE projection of patient embeddings, colored by age and shaped by sex, exhibiting clear demographic structure and smooth age ordering. Cluster densities also correlate with the five clinical sites, indicating that ϕ_p implicitly encodes both demographic and institutional variations without explicit supervision, which is beneficial for personalized federated adaptation.

Impact of Temporal Residual Aggregation Strategy The temporal aggregation strategy via meta-learning dynamically balances contributions from different temporal checkpoints. By assigning adaptive weights to client updates from different time points, our model effectively mitigates temporal concept drift and selectively emphasizes critical clinical events over routine findings.

Impact of Residual-based Update Our residual-based update mechanism inherently stabilizes training under heterogeneous federated conditions. By ensuring each global update remains within a convex combination of previous states, the model prevents large disruptive updates, thereby maintaining stable convergence behavior and avoiding performance degradation often observed in traditional federated algorithms. This stability is empirically reflected by lower variance in validation loss across communication rounds, demonstrating robustness and rapid convergence.

5. Conclusion and Discussion

In this work, we introduced Federated Temporal Adaptation (FTA) as a new federated learning setting that explicitly models both cross-client heterogeneity and temporal evolution in longitudinal medical data. To address this setting, we proposed FedTAR, a parameter-efficient multimodal framework that couples demographics-aware LoRA personalization with meta-learned temporal residual aggregation to handle non-stationary patient trajectories. Experiments on large-scale, multi-institutional datasets show that FedTAR consistently outperforms strong FL baselines in terms of linguistic accuracy, temporal consistency, and convergence stability, while preserving data privacy. Beyond empirical gains, our analysis provides convergence guarantees under mild assumptions, suggesting a principled foundation for extending FTA to irregular visit patterns and more challenging heterogeneous regimes.

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